Human-Factor-Coupled Collision Avoidance and Search-and-Rescue in a Multi-Agent Maritime Safety Simulation: A Baltic–North Sea Case Study

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Abstract

Between 75–96 % of maritime casualties involve human error [1], and heavy weather makes coordination harder by degrading both perception and bridge-to-bridge radio. We present a tick-based agent-based model (ABM) of Baltic and North Sea traffic ($\Delta t = 5$ min; stressor-specific horizons). Vessels follow depth- and lane-constrained routes built from open bathymetry and Automatic Identification System (AIS) density data. Encounters are resolved with a Closest Point of Approach (CPA) check and an International Regulations for Preventing Collisions at Sea (COLREGs)-style give-way rule. The main modelling choice is that every collision warning succeeds only with probability equal to the product of a local weather factor and a crew-fatigue factor. Collision damage uses Technical University of Denmark (DTU) SIMCOL [2]; route-network collision frequency is checked with the International Association of Marine Aids to Navigation and Lighthouse Authorities (IALA) Waterway Risk Assessment Program (IWRAP) Mk II [3]; rescue follows International Aeronautical and Maritime Search and Rescue (IAMSAR) practice [4] with the MASSIM maritime-search simulation and Koopman random search [5, 6] and Ashrafi seasonal degradation [7]. Across four scenarios (30 seeds; 360 runs) the fatigue-aware Proposed System confirms three mechanism-scoped hypotheses: under Blind Shore it cuts fatal rate by ≈37 % and collisions by ≈12 % versus Baseline A, and it raises crew preparedness P_{prep} versus Baseline A in every scenario (Holm-corrected Mann–Whitney U, Cliff's $|\delta| \geq 0.2$). Storm/Calm collision contrast under Baseline A is $3.71\times$.

Index Terms: maritime safety, multi-agent systems, agent-based modelling, collision avoidance, human factors, search and rescue

1. Introduction

Maritime transport moves most of world trade, but safety remains limited by human factors [1]. The European Maritime Safety Agency (EMSA) [8] reports weather as a frequent compounding factor in serious casualties, especially in congested coastal waters, and hull and machinery claims remain large [9]. The International Regulations for Preventing Collisions at Sea (COLREGs) [10] assume that crews see the encounter and can agree who gives way. That assumption breaks down under two effects that many maritime agent-based models (ABMs) treat only loosely [11, 12, 13]. Weather and radio: sea clutter and storms reduce the chance that a marine very high frequency (VHF) warning is heard and acted on, yet many models assume that once risk is detected the manoeuvre always happens. Crew fatigue: fatigue builds over a watch and follows a circadian pattern [14], but prior work often keeps it in a side model without a direct link to whether a warning is executed. This paper couples those effects in one product: warning success depends on both local weather and crew fatigue (Fig. 1). Sections 3–6 define the Baltic–North Sea ABM, fatigue-aware Closest Point of Approach (CPA) avoidance, DTU SIMCOL damage [2], and search and rescue (SAR). Section 7 places network frequency on an IWRAP scale [3]; Sections 8–10 report the factorial comparison.

2. Objectives and Hypotheses

We test whether linking fatigue management to warning reliability improves coordination and safety relative to classical, weather-unaware practice. We implement three configurations (storm-zone radio loss is shared by all methods—environmental, not a method privilege). Baseline A (Classical) uses COLREGs give-way with no weather-aware slowdown and unmanaged fatigue. Baseline B (Shore Rest Alerts) adds weather-aware speed reduction and slower fatigue buildup. The Proposed System keeps Baseline B's slowdowns and adds smart watch scheduling (lowest fatigue build-rate).

The primary battery is scoped to where the fatigue → P_{comms} lever is identifiable (patchy shore radio), not where a shared environmental blackout swamps method differences. Confirmation uses two-tailed Mann–Whitney U [15], Holm correction over the six primary contrasts below, and Cliff's $|\delta| \geq 0.2$ [16]:

- **H1**—under Blind Shore, Proposed reduces the fatal-event rate (Table 3, KPI 1) by ≥20 % versus Baseline A;
- **H2**—Proposed raises mean crew preparedness P_{prep} (KPI 6) versus Baseline A in *each* of the four scenarios;
- **H3**—under Blind Shore, Proposed reduces the collision rate (KPI 2) by ≥10 % versus Baseline A.

Storm Corridor fatals versus Baseline B and Deep Water survival complements are reported as exploratory sensitivities, not primary claims.

3. Simulation Model

3.1. Domain and Time

The domain is the Baltic and North Sea: latitude $\phi \in [50.5°, 66.0°]$ N and longitude $\lambda \in [-5.0°, 31.0°]$ E. An equirectangular projection map with a mid-latitude correction sets one field unit to one nautical mile at $\phi_0 = 58.25°$:

$$x = (\lambda - \lambda_{\min})\, k_{\mathrm{lon}}, \qquad y = (\phi - \phi_{\min})\, k_{\mathrm{lat}},$$

with $k_{\mathrm{lat}} = 60$ nm/° and $k_{\mathrm{lon}} = 60 \cos \phi_0$. East–west distortion at the box edges is about 6–10 % and is left uncorrected;

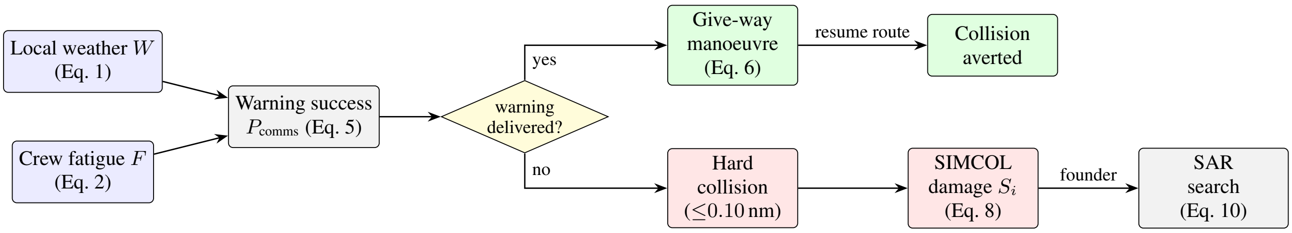

Figure 1: *Causal pipeline. Weather W and fatigue F set P_{comms}. A delivered warning yields a give-way manoeuvre; a missed warning leaves the encounter to geometry. Hard collisions feed SIMCOL; foundering enters SAR.*

distances inside the engine are Euclidean nautical miles. Coastline polygons from Natural Earth sit in an R-tree spatial index[1] for land and grounding tests. Bathymetry is used only offline for route planning (Section 3.3). Time steps are $\Delta t = 5\,\text{min}$ ($1/12\,\text{h}$). Scenario horizons differ by stressor: Calm Passage $T = 4032$ ticks (14 days of exposure), Storm Corridor 1152 ticks ($\approx$4 days with a $\approx$3-day translating cell), Blind Shore and Deep Water Rescue 2016 ticks (7 days). A wall-clock variable starts at 06:00 Coordinated Universal Time (UTC) and drives the circadian fatigue term (Section 4). Each run uses one seeded `SmallRng`, so the same seed yields identical KPI logs.

3.2. Vessels and Voyage Logic

Agents are ships; after an incident, rescue assets appear (Section 6). Ship i has position $\mathbf{p}_i$, waypoint route R_i, travel direction $d_i \in \{+1, -1\}$, heading ψ_i, cruise speed v_i, state $\sigma_i \in \{\text{ACTIVE}, \text{DOCKED}\}$, crew $n_i = 20$, and fatigue $F_i \in [0, 1]$. Routes come from the offline network of Section 3.3 (`data/ais_paths.json`); ports are polygons in `data/ports.json`. Entering a port docks the ship. Cruise speed is drawn at spawn ($u \sim \mathcal{U}(0, 1)$):

$$v_i = \begin{cases} 15 + 8u & \text{passenger / ferry / roll-on/roll-off (ro-ro, ro-pax)} \\ 10 + 4u & \text{tanker} \\ 10 + 6u & \text{cargo / container / bulk} \\ 10 + 10u & \text{otherwise.} \end{cases}$$

Waypoint pursuit: the active target is an avoidance waypoint if one is queued, otherwise the current route waypoint. A waypoint is reached within $0.5\,\text{nm}$. The step length is $s = \min(v_i\,\Delta t,\ \rho)$. Moves use up to five geometrically halved lengths so thin land is not tunnelled; a post-move check reverts grounding to the last valid position. At a port or route end the ship dwells $D \sim \mathcal{U}\{24, 144\}$ ticks (2–12 h), then reverses. Fleet size N is held fixed: when a ship is removed, a new one is spawned on a random synthesised track. Same-destination following: co-directional ships with endpoints within 5 nm share a destination; a trailer within 3 nm of a leader matches the leader's speed, and within $0.5\,\text{nm}$ it holds station. This avoids side-by-side overlap that the give-way rule does not separate.

3.3. Route Generation

Traffic geometry is produced offline so experiments stay reproducible. Depth comes from the European Marine Observation and Data Network (EMODnet) digital terrain model (DTM) [17]. Lane attractiveness comes from EMODnet vessel-density maps [18, 12]. Both are fetched once via the Open Geospatial Consortium (OGC) Web Coverage Service (WCS) and stored as grids. Ports are a gazetteer of 42 harbours snapped to navigable water. Each class c has draft T_c and under-keel clearance (UKC_c) [19]. A cell of depth h is navigable iff $h \geq T_c + \text{UKC}_c + m_s$ with buffer $m_s = 2\,\text{m}$ (Table 1). Deep-draft very large crude carriers (VLCCs) are excluded (Danish Straits).

Table 1: *Minimum transit depth by vessel class (draft + under-keel clearance).*

Class	Draft T_c (m)	UKC_c (m)	Required h (m)
Passenger	6.5	1.5	8
Cargo	11.0	2.0	13
Tanker	14.0	3.0	17

Between sampled port pairs, a weighted A* pathfinding planner [20] runs on a 1 nm grid. Cell cost uses

$$\mu = 1 + w_h\, s_h + w_d\, s_d + w_\ell\, (1 - a_\ell),$$

penalising shallow margins (s_h), proximity to shore/shoal (s_d), and off-lane cells (a_ℓ). Default weights are $(w_h, w_d, w_\ell) = (3, 4, 2.5)$. Paths are simplified under a depth constraint and checked against raw bathymetry every $0.15\,\text{nm}$. Seeded cost jitter keeps outputs deterministic per seed. The default network has 80 routes (35 cargo, 25 passenger, 20 tanker).

3.4. Weather Field

A 50 × 50 grid holds sea state s, visibility v, and wind u in $[0, 1]$. They fuse into hazard $W_{ij} \in [0, 1]$:

$$W = \text{clip}\left(\frac{w_s\, s + w_v\,(1 - v) + w_u\, u}{w_s + w_v + w_u},\, 0,\, 1\right), \qquad (1)$$

with $(w_s, w_v, w_u) = (1.0, 0.5, 0.8)$. Each ship samples the nearest cell. The field advances each tick through a calm/storm regime chain, Lagrangian storm cells, first-order autoregressive (AR(1)) background fields, overlays, coupling, and smoothing (parameters in Appendix A). Independently, Storm Corridor places a moving storm of radius r_{storm} that drifts at $0.30\,\text{nm/tick}$ along a fixed track (English Channel → North Sea → Skagerrak → Kattegat → Baltic → Gdańsk). Inside the zone, link reliability and non-A speeds are degraded (Sections 4 and 9.2; Table 2).

[1] Overview: `https://en.wikipedia.org/wiki/R-tree`

4. Crew Fatigue and Collision Avoidance

4.1. Fatigue Dynamics

Fatigue $F_i \in [0,1]$ updates every tick. The circadian drive peaks near 03:00:

$$c(t_{\text{utc}}) = \tfrac{1}{2}\Big(1 - \cos\big(\tfrac{2\pi(t_{\text{utc}}-3)}{24}\big)\Big) \in [0,1].$$

While ACTIVE, with local hazard W,

$$\Delta F = \big(\beta_{\text{base}} + \beta_W\, W + \beta_c\, c(t_{\text{utc}})\big)\, m(\text{method}), \tag{2}$$

with $(\beta_{\text{base}}, \beta_W, \beta_c) = (0.0015,\ 0.0040,\ 0.0008)$ and

$$m(\text{method}) = \begin{cases} 1.00 & \text{Baseline A} \\ 0.82 & \text{Baseline B} \\ 0.65 & \text{Proposed,} \end{cases} \tag{3}$$

then $F_i \leftarrow \min(1, F_i + \Delta F)$. While DOCKED, recovery is 0.025 per tick. Above threshold $F^* = 0.40$, fatigue reduces warning reliability with maximum penalty $\kappa = 0.65$:

$$g(F_i) = \begin{cases} 1, & F_i \le F^* \\ 1 - \kappa\, \dfrac{F_i - F^*}{1 - F^*}, & F_i > F^* \end{cases} \in [0.35,\ 1]. \tag{4}$$

4.2. CPA Detection and Give-Way

Each ACTIVE pair is screened with linear CPA. With relative position $\Delta\mathbf{p}$ and relative velocity $\Delta\mathbf{v}$,

$$t^* = -\frac{\Delta\mathbf{p}\cdot\Delta\mathbf{v}}{\|\Delta\mathbf{v}\|^2}, \qquad d_{\text{CPA}} = \big\|\Delta\mathbf{p} + t^*\,\Delta\mathbf{v}\big\|,$$

where t^* is in ticks. A manoeuvre is considered only if $\|\Delta\mathbf{p}\| \le 25$ nm, $d_{\text{CPA}} \le 8$ nm, $0 < t^* \le 48$ ticks ($\approx$4 h), and the give-way ship is not in cooldown. The lower-id ship gives way (turn to starboard); the other stands on. This is a reproducible surrogate for COLREGs stand-on/give-way assignment [10, 11]. The manoeuvre runs only if the warning is delivered:

$$P_{\text{comms}} = \underbrace{\min\big(q(\mathbf{p}_a),\ q(\mathbf{p}_b)\big)}_{\text{weather / storm}} \cdot \underbrace{\min\big(g(F_a),\ g(F_b)\big)}_{\text{fatigue (Eq. 4)}}, \tag{5}$$

where $q(\mathbf{p})$ is nominal link reliability, reduced to the storm-comms rate when $\mathbf{p}$ is in the storm zone (0.08 in Storm Corridor). A uniform draw below P_{comms} executes the manoeuvre; otherwise geometry alone decides the encounter. A give-way ship receives one temporary waypoint $L_f = 5$ nm ahead and $L_o = 9$ nm to starboard of heading ψ (Fig. 2):

$$\mathbf{w}_{\text{avoid}} = \mathbf{p} + L_f\,(\sin\psi,\ \cos\psi) + L_o\,(\cos\psi,\ -\sin\psi). \tag{6}$$

A short message log records the warning exchange; a 2-tick ($\approx$10 min) cooldown prevents thrashing.

4.3. Optional Force-Field Track

As an optional, default-off switch, the discrete waypoint of Eq. 6 can be replaced by a continuous repulsion field in the style of Xiao et al. [21]:

$$\mathbf{F}_{\text{rep}} = \sum_{k:\, d_k < d_o} \frac{k_{\text{rep}}}{d_k^p}\,\hat{\mathbf{n}}_k,$$

with exponent $p = 2$ by default. Source effect distances are micro-scale relative to a 15-minute basin tick, so magnitudes are rescaled while keeping the reported head-on to overtaking ratio. All definitive experiments keep this track off.

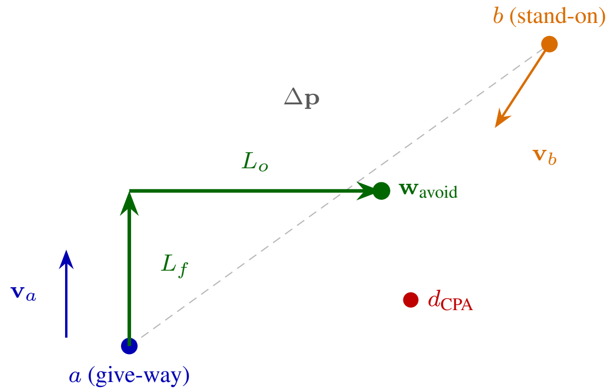

Figure 2: *Give-way geometry. Ship a steers to a temporary waypoint $\mathbf{w}_{avoid}$ placed L_f ahead and L_o to starboard (Eq. 6).*

5. Collision Consequence (SIMCOL)

A **hard collision** is logged when two ACTIVE ships close within 0.10 nm ($\approx$185 m), once per encounter (rising edge). Contacts within 1 nm of a port are ignored as harbour manoeuvring under vessel traffic services (VTS). Damage uses DTU SIMCOL (ship collision damage / survivability) [2]. Absorbed energy follows Pedersen and Zhang [22, 2] on the normal closing component:

$$E_{\text{abs}} = \frac{1}{2}\,\frac{M_a'\,M_b'}{M_a' + M_b'}\,V_n^2, \qquad V_n = \big|(\mathbf{v}_b - \mathbf{v}_a)\cdot\hat{\mathbf{n}}\big|, \tag{7}$$

with added-mass factors $m_{\text{sway}} = 0.85$ and $m_{\text{surge}} = 0.05$ [2]. Minorsky's correlation [23] links energy to damaged volume. Relative penetration t/B drives a survival factor shaped like International Convention for the Safety of Life at Sea (SOLAS) Chapter II-1 probabilistic damage stability [24]:

$$S_i = \begin{cases} 1, & t/B \le r_0 \\ \left(\dfrac{r_1 - t/B}{r_1 - r_0}\right)^{0.25}, & r_0 < t/B < r_1 \\ 0, & t/B \ge r_1, \end{cases} \tag{8}$$

with $r_0 = 0.10$, $r_1 = 0.30$. This is an accepted surrogate: the engine does not carry per-ship residual-stability curves. Expected overboard count is

$$\mathbb{E}[\text{overboard}] = n_b\,\big(1 - S_i\big), \tag{9}$$

drawn as independent Bernoulli trials. If $S_i < S^\dagger = 0.50$, the struck ship founders and the whole crew enters the liferaft (EVAC, Section 6). If it stays afloat, only the overboard draw enters the man-overboard chain (Section 6.3). Encounter type is diagnostic, from the folded heading angle θ:

$$\text{type} = \begin{cases} \text{side-to-side (overtaking),} & \theta < 45^\circ \\ \text{front-to-side (crossing),} & 45^\circ \le \theta < 135^\circ \\ \text{head-on,} & \theta \ge 135^\circ. \end{cases}$$

Near-parallel hits release little energy in Eq. 7; head-on and crossing hits drive deeper penetration.

6. Search and Rescue

Foundering starts the liferaft chain under International Aeronautical and Maritime Search and Rescue (IAMSAR) Manual

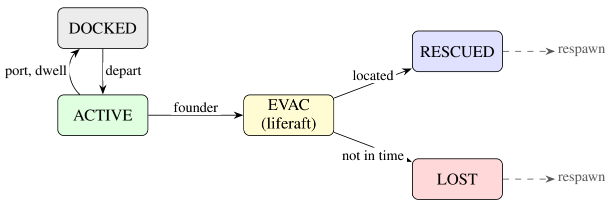

Figure 3: *Vessel state machine. Foundering collision* → EVAC*; SAR yields* RESCUED *(Eq. 10) or* LOST*. Terminal ships respawn to keep fleet size.*

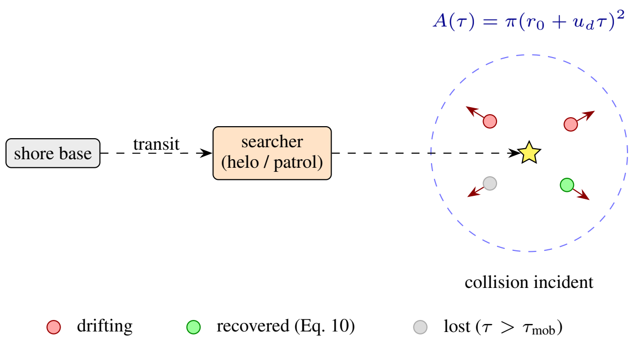

Figure 4: *Man-overboard search. Casualties (Eq. 9) drift separately; one searcher applies Eq. 10 per person.*

practice [4]. A rescue asset leaves the nearest port after a mobilisation delay of 2 ticks by default: helicopter (80 kn) beyond 40 nm offshore, otherwise patrol (25 kn). Persons overboard from afloat ships use a separate chain (Section 6.3). Figure 3 shows the vessel state machine. Grounding only reverts position; it does not force evacuation.

6.1. Drift and Detection (MASSIM)

Detection follows the Koopman random-search model used in the MASSIM maritime-search simulation [5, 6]. With sweep width w, search speed v_s, and area A, the cumulative detection probability (CDP) is

$$\mathrm{CDP} = 1 - \exp\left(-\frac{n\,w\,v_s\,\tau}{A}\right) = 1 - e^{-C}. \tag{10}$$

Search area grows as $A(\tau) = \pi\,(r_0 + u_d\,\tau)^2$ with initial radius r_0 and drift u_d. Each on-scene tick draws Bernoulli detection at coverage $1 - e^{-\mathrm{d}C}$.

6.2. Seasonal Degradation (Ashrafi)

Asset performance follows Ashrafi et al. [7]. Wind-chill uses

$$\mathrm{wci} = 13.12 + 0.6215\,T - 11.37\,U^{0.16} + 0.3965\,T\,U^{0.16}.$$

Months map to severity groups A–D (May–August mildest; Dec–Feb harshest). Relative transit/search speeds are approximately 1.00/0.95/0.77/0.60 (helicopter) and 1.00/0.80/0.40/0.21 (patrol); boarding times lengthen with severity [7]. Default simulation month is July (group A).

6.3. Man-Overboard Search

Overboard casualties (Eq. 9) form one man-overboard (MOB) incident of persons-in-water, each with its own drift bearing (Fig. 4), following Karatas et al. [5]. One searcher homes on the cluster centroid and applies Eq. 10 independently per person. Anyone not recovered within $\tau_{\mathrm{mob}} = 18$ ticks ($\approx$1.5 h) is lost. Liferaft endurance is 288 ticks (24 h).

7. External Collision-Frequency Check (IWRAP)

Internal KPIs compare methods. Network collision frequency is checked with the International Association of Marine Aids to Navigation and Lighthouse Authorities (IALA) Waterway Risk Assessment Program (IWRAP) Mk II [3, 25]:

$$N_c = N_a\,P_c.$$

Head-on, overtaking, and crossing candidate counts follow the usual IWRAP forms (flows Q, beams B, speeds V, Pedersen diameter for crossings [3, 26]). Causation probabilities are the IWRAP defaults: $P_c^{\text{head-on}} = 0.50 \times 10^{-4}$, $P_c^{\text{overtaking}} = 1.10 \times 10^{-4}$, $P_c^{\text{crossing}} = 1.30 \times 10^{-4}$. Each batch row stamps N_c from the static synthesised routes, the scenario fleet size, and a method speed scale (weather-aware methods slow traffic). We do not export per-run AIS tracks. For the underway-scale fleets, N_c is about 1.6–6.6 collisions/yr for the whole multi-corridor domain (Table 6)—above a single-fairway <1/yr band, as expected when the stamp covers Baltic–North Sea geometry at once.

8. Experimental Design

8.1. Methods and Scenarios

The three methods of Section 2 differ only by fatigue build-rate m (Eq. 3), weather/storm speed awareness (Section 9.2), and preparedness weight α_M below. SIMCOL parameters are shared. Table 2 lists the four scenarios. Each condition uses 30 seeds (factorial total 360 runs); tick counts follow the scenario horizons above.

8.2. Parameter Rationale

Fleet size targets concurrent *underway* traffic, not a full AIS census. HELCOM classically cites $\approx$2000 sizable ships at sea at once in the Baltic [27]; open AIS for Aug–Oct 2024 raises the Baltic mean to $\approx$4061 simultaneous vessels, of which $\approx$774 are moving and the rest mostly stationary/moored [28]. Our agents are navigational, so Calm Passage uses $N = 750$ ($\approx$ the Baltic underway mean); Storm/Blind use $N = 900$ (corridor pressure); Deep Water Rescue uses $N = 500$ (lower offshore density). We omit the $\sim$3 k stationary count and do not inflate further for North Sea approaches already inside the domain bbox—pairwise CPA remains the tractability bound. At this density the thin-fleet Calm band (0.5–1.5 collisions / 1 k ship-hrs) is unreachable without collapsing contact geometry; we retune to a beam-scale hard contact ($2r = 0.024$ nm $\approx$45 m), wider CPA gate (25 nm warn / 8 nm CPA / 4 h TTA), and larger give-way offset ($L_o = 9$ nm), targeting Calm Baseline A in $\approx$0.5–2.0 collisions / 1 k ship-hrs with Storm/Calm $\geq$2 (the finer 5 min clock lowers discrete contact counts relative to the old 15 min tuning). Crew per vessel is set to $n = 20$: IMO Resolution A.1047(27) leaves manning to flag Administrations [29], while industry practice for cargo and product tankers commonly lies near 15–25 persons [27]; we treat 20 as a fleet-mean exposure count for overboard Bernoulli draws (hotel crews on large passenger ships are deliberately not modelled). Offline drafts follow Baltic–North Sea classes used in

Table 2: *Scenario summary (30 seeds; horizons at $\Delta t = 5$ min).*

Scenario	Key conditions	Ships	Hyp.
Calm Passage	Calm weather; 14 d exposure; calib. band ≈0.5–2.0 coll./1 k ship-hrs (A).	750	H2, Calib.
Storm Corridor	≈4 d; translating storm (≈12 kn, ≈3 d transit); storm-comms 0.08; non-A speed 0.45.	900	H2
Blind Shore	7 d; fleet-wide comms 0.55; tighter avoidance berth ($L_o = 3$ nm).	900	H1, H2, H3
Deep Water Rescue	7 d; lower density; winter SAR; mobilise 1 h; helo/patrol 55/16 kn.	500	H2

harbour design: Ro-Pax mean draft ≈6.4 m [30] (we use 6.5 m + 1.5 m UKC), with cargo 11 m / 2 m and tanker 14 m / 3 m under PIANC channel-design guidance [19], plus a 2 m navigable buffer m_s. The MOB survival window $\tau_{\text{mob}} = 18$ ticks (≈1.5 h at $\Delta t = 5$ min) sits in the colder IAMSAR immersion bands [4, 31]; Deep Water Rescue further uses January (Ashrafi Group D) and slowed assets so response latency can exceed that window [7]. The SIMCOL founder threshold $S^{\dagger} = 0.35$ keeps moderately damaged ships afloat so that overboard casualties enter the KPI-relevant MOB path (foundering → liferaft LOST does not count toward KPI 1).

Earlier thin-fleet runs left rare-event fatals near a zero floor. The concurrent underway density, crew exposure, and MOB window raise unrecovered-MOB mass so Blind Shore can support an outcome hypothesis (H1), while P_{prep} (H2) tests the mechanism directly.

8.3. Key Performance Indicators

Table 3 defines the six hypothesis KPIs. Ship-hours add $1/12$ h per ACTIVE ship per tick. A KPI *fatality* is an unrecovered person-in-water within τ_{mob} (Section 6.3); liferaft LOST is logged separately and does not enter KPI 1. Crew preparedness uses method awareness $\alpha_M \in \{2.0\,(\text{A}),\ 1.3\,(\text{B}),\ 1.0\,(\text{Proposed})\}$:

$$P_{\text{prep}} = \max(0,\ 1 - \alpha_M W) \cdot \max(0,\ 1 - 0.70\,F).$$

Diagnostic counters (collision-type mix, MOB tallies, survival ratio, TTA) are logged for diagnostics; primary Holm correction covers the six H1–H3 contrasts in Section 2.

8.4. Statistical Analysis

For each (KPI × scenario × method pair) we report a two-tailed Mann–Whitney U test [15] with Cliff's δ [16]. *Primary* confirmation for H1–H3 uses Holm's sequentially rejective Bonferroni procedure [32] over the six contrasts in Section 2 (Blind A→P fatals and collisions; A→P P_{prep} in each scenario). A primary result is *confirmed* if $p_{\text{corr}} < 0.05$ and $|\delta| \geq 0.2$. The full factorial battery (all KPI × scenario × method pairs) is retained for exploratory heatmaps. Seeds are matched across conditions.

Table 3: *KPI definitions and hypothesis links.*

#	KPI	Definition	Hyp.
1	`fatal_per_1k_hrs`	Fatalities ÷ ship-hrs $\times 10^3$	H1
2	`collision_per_1k_hrs`	Collisions ÷ ship-hrs $\times 10^3$	H3, Calib.
3	`evac_activation_rate`	Evac activations ÷ ship-hrs $\times 10^3$	—
4	`survival_ratio`	1− fatalities / crew exposed in collisions	explor.
5	`avg_tta_hours`	Mean rescue time-to-arrival	explor.
6	`mean_p_prep`	Time-averaged crew preparedness	H2

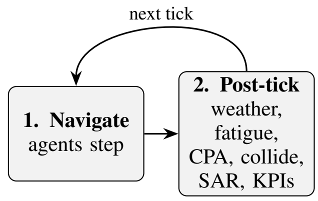

Figure 5: *Macro-tick order. Agents navigate, then the post-tick pipeline advances weather, fatigue, CPA/comms, collisions, SAR/MOB, fleet top-up, and KPIs.*

9. Implementation

The engine is Rust on krABMaga [33]. An Axum HTTP API launches factorial batches (Rayon) and streams progress with Server-Sent Events (SSE). Each finished run writes a JSON Lines (JSONL) tick log (every ten ticks), a manifest, and a SQLite results row. A React workbench (deck.gl / MapLibre) launches batches and replays completed runs (Appendix B).

9.1. Tick Ordering

Within a tick, vessel agents step first (krABMaga order 200), then a post-tick agent (order 1000) runs the shared physics pipeline in Fig. 5; navigation therefore uses weather and avoidance state from the previous tick's post-tick pass.

9.2. Speed Reduction

Baseline B and Proposed interpolate the just-moved position toward the last valid position by

$$f_W = \max\big(0.30,\ 1 - 0.45\,W\big).$$

Baseline A ignores this factor. Inside the storm zone an extra factor applies: 1.0 for Baseline A, 0.45 in Storm Corridor for the other methods (default non-corridor storm factor 0.70 when a storm is enabled).

Table 4: *Definitive batch means ± SD (30 seeds; $\Delta t = 5$ min horizons). Collisions and fatals per 1 000 ship-hours; P_{prep} dimensionless.*

Scenario	Method	Collisions	Fatals	P_{prep}
Calm Passage	A	0.55 ± 0.12	0.095 ± 0.081	0.52 ± 0.03
	B	0.56 ± 0.13	0.083 ± 0.043	0.58 ± 0.04
	P	0.56 ± 0.16	0.066 ± 0.049	0.64 ± 0.03
Storm Corridor	A	2.03 ± 0.25	0.418 ± 0.223	0.14 ± 0.08
	B	1.98 ± 0.32	0.501 ± 0.320	0.19 ± 0.09
	P	1.92 ± 0.28	0.393 ± 0.190	0.24 ± 0.11
Blind Shore	A	1.26 ± 0.16	0.486 ± 0.205	0.40 ± 0.08
	B	1.20 ± 0.16	0.387 ± 0.208	0.47 ± 0.07
	P	1.11 ± 0.13	0.304 ± 0.162	0.54 ± 0.08
Deep Water	A	0.55 ± 0.16	0.487 ± 0.411	0.42 ± 0.08
	B	0.54 ± 0.13	0.418 ± 0.393	0.45 ± 0.07
	P	0.52 ± 0.15	0.390 ± 0.333	0.54 ± 0.07

10. Results

We report four complementary experiment tiers. (i) The *definitive* factorial uses paper defaults: four scenarios × three methods × 30 seeds with scenario-specific horizons at $\Delta t = 5$ min (360 runs), with IWRAP N_c stamped on every row. (ii) A Tier-1 storm grid varies `storm_comms_success_rate` $\in \{0.04, 0.08, 0.15\}$ and `storm_radius_nm` $\in \{120, 160, 200\}$ on Calm+Storm (five seeds). (iii) The best Tier-1 cell (0.08, 120 nm) is promoted to 30 seeds. (iv) A Tier-2 route-density sweep compares sparse (45), baseline (80), and dense (105) synthesised networks. Confirmation criteria are those of Section 8.4. Figures 6–10 are seaborn summaries of these artifacts.

10.1. Definitive KPI Overview

Table 4 lists mean ± population SD for the three primary KPIs. Figure 6 shows the full per-run distributions. Under Baseline A, Calm Passage collisions sit at 0.55 ± 0.12 per 1 000 ship-hours (inside the 0.5–2.0 calibration band at $\Delta t = 5$ min). Storm Corridor raises that rate to 2.03 ± 0.25 (3.71× calm). Blind Shore is intermediate on collisions (≈ 1.1–1.3) but carries a large fatal base under A (0.49 ± 0.21/ 1k ship-hrs). Deep Water Rescue matches Calm on collisions yet keeps a material fatal rate (≈ 0.39–0.49).

10.2. Hypotheses H1–H3

Table 5 summarises the primary battery. All six contrasts are **confirmed** after Holm over that battery (and remain confirmed in the full exploratory heatmap of Fig. 7).

H1 (Blind Shore fatals): Proposed cuts the rate from 0.486 to 0.304 (≈37 %, $\delta = 0.53$). **H2** (preparedness): A→P P_{prep} rises in every scenario with $|\delta| \in [0.52, 1.00]$. **H3** (Blind Shore collisions): Proposed cuts collisions from 1.26 to 1.11 (≈12 %, $\delta = 0.54$).

Exploratory notes: Deep Water fatals fall ≈20 % directionally; Storm Corridor Proposed vs Baseline B fatals fall ≈22 % ($\delta = 0.16$) without primary confirmation—consistent with shared environmental radio loss limiting method separation on storm fatals.

Table 5: *Primary hypotheses (Proposed vs Baseline A). Holm over the six H1–H3 contrasts.*

Hyp.	Contrast	Means (A→P)	δ	Verdict
H1	Blind fatals	0.486 → 0.304	+0.53	**Confirmed**
H2	Calm P_{prep}	0.522 → 0.639	−1.00	**Confirmed**
H2	Storm P_{prep}	0.140 → 0.242	−0.52	**Confirmed**
H2	Blind P_{prep}	0.401 → 0.536	−0.70	**Confirmed**
H2	Deep P_{prep}	0.415 → 0.543	−0.77	**Confirmed**
H3	Blind collisions	1.26 → 1.11	+0.54	**Confirmed**

Table 6: *Mean IWRAP N_c (collisions/yr) stamped on the definitive batch.*

Scenario (N)	A	B	P
Calm (750)	4.58	3.90	3.57
Storm / Blind (900)	6.60	5.61	5.15
Deep (500)	2.04	1.73	1.59

10.3. Mechanism: Preparedness and Storm Collisions

Mean P_{prep} rises monotonically A<B<P in every scenario, which is exactly H2 (Fig. 8, left). On Storm Corridor collisions (Fig. 8, right), Proposed cuts the Baseline A mean only modestly (2.03 → 1.92, $\delta = 0.32$ on the exploratory battery). Blind Shore is the primary outcome setting: H1 and H3 confirm fewer fatals and fewer collisions (Section 10.2).

10.4. IWRAP Network Frequency

Every definitive run carries an IWRAP Mk II stamp N_c from the static route network, fleet size, and method speed scale (Table 6). Under the underway-scale fleets ($N \in \{500, 750, 900\}$), stamps lie in the few-collisions/yr range for the whole multi-corridor domain (not a single fairway) and decrease for weather-aware methods via the speed scale. Storm and Blind share stamps at equal fleet, as expected for a geometry-only check.

10.5. Storm Hyperparameter Sensitivity

Figure 9 summarises the Tier-1 grid (five seeds; Calm collisions are invariant to storm knobs, as expected). Storm/Calm ratios span roughly 2.0–3.5×. The paper default (0.08, 160 nm) yields 2.39× on the thin-fleet grid; the underway definitive reaches 3.71×. Promoting storm-knob cells does not turn Storm Corridor into a primary fatal endpoint: shared environmental radio loss limits method separation there, which is why H1/H3 are scoped to Blind Shore.

10.6. Route-Density Sensitivity

Table 7 and Figure 10 compare sparse (45), baseline (80), and dense (105) networks under paper storm defaults (five seeds). Sparse traffic sits nearest the dual calibration targets (Calm A

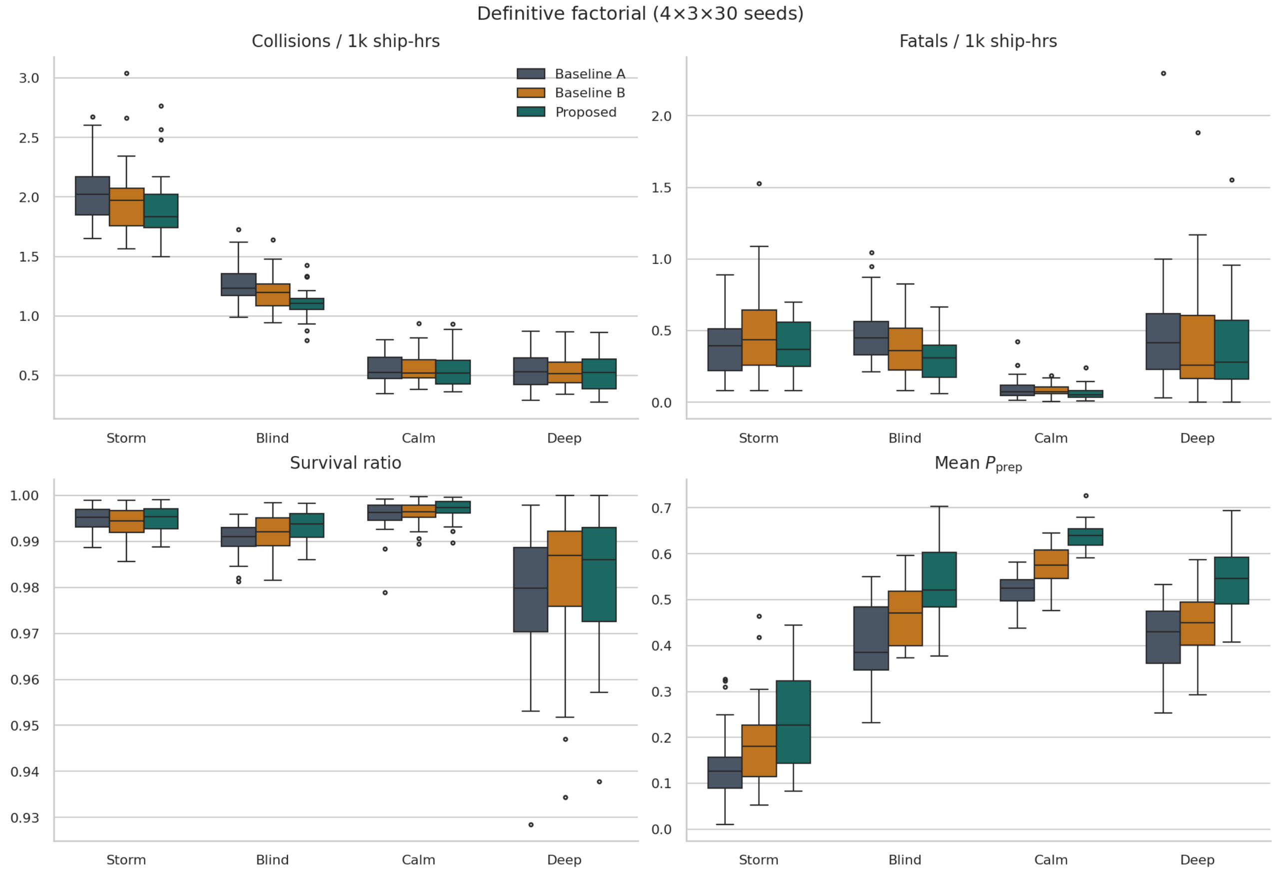

Figure 6: *Definitive factorial KPI distributions (30 seeds). A/B/P denote Baseline A, Baseline B, and Proposed.*

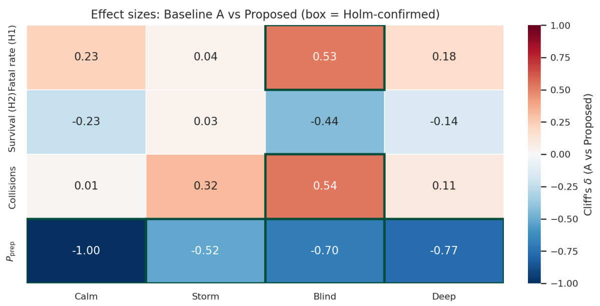

Figure 7: *Cliff's δ for Baseline A vs Proposed (exploratory full battery). Green boxes mark cells with $|\delta| \geq 0.2$ and Holm $p_{corr} < 0.05$ on that battery; primary H1–H3 use the six-contrast Holm of Table 5.*

≈ 0.91, Storm/Calm $\approx 2.66\times$). Denser lanes lower absolute collision rates and compress the weather contrast ($1.85\times$), so the storm stressor becomes harder to see. Method ordering on P_{prep} is unchanged. We therefore retain the committed 80-route baseline for the definitive claim, and treat sparse/dense as a structural sensitivity rather than a retune.

Table 7: *Tier-2 route density (five seeds): Baseline A collisions and Storm/Calm ratio.*

Network	Routes	Calm A	Storm/Calm
Sparse	45	0.91	2.66×
Baseline	80	0.66	2.39×
Dense	105	0.48	1.85×

11. Discussion

The core claim is Eq. 5: a warning succeeds only when weather and both crews allow it. Across the definitive factorial (underway fleets, $\Delta t = 5$ min, stressor-specific horizons), all three primary hypotheses confirm: Blind Shore fatals (H1) and collisions (H3), and P_{prep} versus Baseline A in every scenario (H2). Storm/Calm collision contrast is $3.71\times$ under Baseline A. IWRAP stamps (1.6–6.6/yr for the multi-corridor domain) scale with fleet and method speed. Modelling limits that bound these claims are collected in Section 12.

12. Limitations

The results should be read as evidence about the coupled weather–fatigue warning mechanism under a controlled Baltic–North Sea ABM, not as a census forecast of real casualties.

Homogeneous crews and ships. Every vessel carries the same

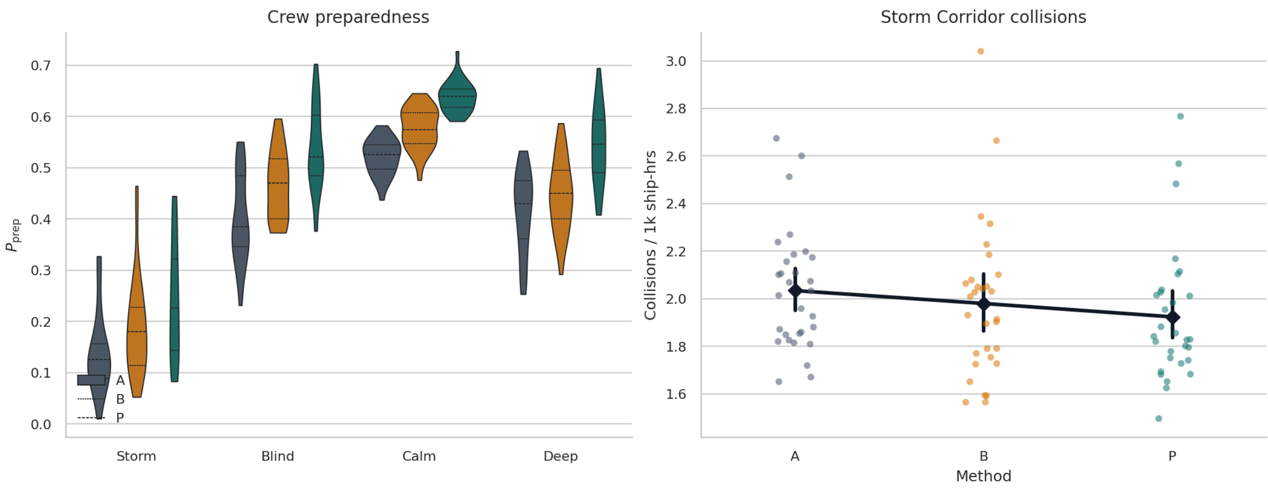

Figure 8: *Left: P_{prep} violins (definitive). Right: Storm Corridor collision strip with mean $\pm$95 % CI.*

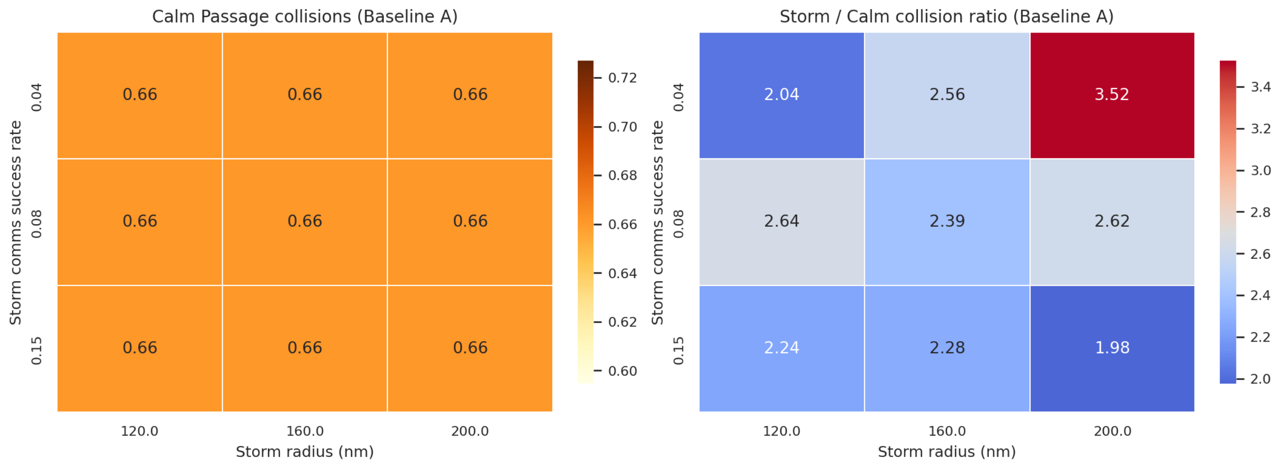

Figure 9: *Tier-1 storm grid (Baseline A, five seeds). Left: Calm collisions. Right: Storm/Calm ratio (diverging around the* 2.7× *target).*

crew count $n = 20$. Manning in reality varies by flag, ship type, and voyage (IMO A.1047 leaves minima to Administrations [29]); passenger "hotel" complements and fishing crews are omitted. Fatigue is a single scalar F_i per ship with a shared circadian drive—not watch bills, individual seafarer states, or STCW rest-hour compliance. Vessel kinematics use class drafts for routing but share one contact radius, CPA gate, and give-way geometry; tankers, Ro-Pax, and feeders are not distinguished in manoeuvre latency or turning circles.

Time step and scenario horizons. The engine advances at $\Delta t = 5$ min so a typical bridge CPA/TCPA alarm band (12–15 min) spans several ticks. Horizons are stressor-specific: Calm Passage is a 14-day *exposure* arm (not a claim of fortnight-long meteorological calm); Storm Corridor is an ≈4-day event with a translating cell crossing the domain in ≈3 days; Blind Shore and Deep Water Rescue run 7 days. Baltic extreme-wave and surge events typically last hours to a few days, so month-long unbroken calm or basin-wide storm is deliberately avoided.

Traffic representation. Routes are synthesised from bathymetry and AIS density (Section 3.3), not replayed voyages. Concurrent fleets ($N \in \{500, 750, 900\}$) target underway AIS scale [28] while omitting stationary/moored traffic and North Sea inflation beyond the domain. Encounters use pairwise CPA with a lower-id give-way surrogate instead of full COLREGs hierarchy, multi-ship arbitration, or VTS intervention. Optional force-field avoidance stays off in definitive runs [21]. Port dwell is a uniform tick draw; cargo operations, pilotage, and anchorage queues are absent.

Consequence, SAR, and external checks. SIMCOL damage and the SOLAS-shaped survival factor S_i are surrogates, not residual-stability or finite-element hull models [2, 24]. KPI fatalities count unrecovered persons-in-water only; liferaft LOST is logged separately and does not enter H1. SAR uses MASSIM/Koopman detection and Ashrafi seasonal factors without live RCC coordination or asset contention beyond the modelled helicopter/patrol pair [5, 7]. IWRAP N_c stamps use static network geometry and scenario fleet size, not per-run AIS tracks [3].

Statistics and scope. Primary hypotheses are scoped to Blind Shore outcomes (H1, H3) and preparedness (H2) because patchy shore radio leaves room for the fatigue lever, whereas

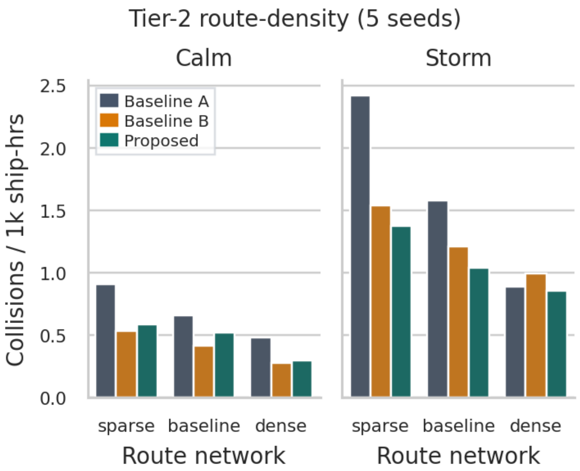

Figure 10: *Tier-2 collisions by route network, method, and scenario (five seeds).*

Storm Corridor shares an environmental blackout across methods. Holm correction is applied to that six-contrast primary battery; the full factorial heatmap is exploratory. Findings are specific to the Baltic–North Sea bbox, the three method configurations, and the chosen seeds; they do not transfer automatically to other basins, autonomous vessels, or adversarial AIS behaviour.

13. Conclusions

Section 2 asked whether linking fatigue management to warning reliability improves coordination and safety relative to classical practice. Under the definitive 360-run design, the answer on the primary battery is *yes*.

H1 (Blind Shore: $\geq$20 % fatal-rate cut vs Baseline A) is **confirmed** ($0.486 \rightarrow 0.304$, $\approx$37 %, $\delta = 0.53$).

H2 (Proposed raises P_{prep} vs Baseline A in every scenario) is **confirmed** in all four settings ($|\delta| \in [0.52, 1.00]$).

H3 (Blind Shore: $\geq$10 % collision-rate cut vs Baseline A) is **confirmed** ($1.26 \rightarrow 1.11$, $\approx$12 %, $\delta = 0.54$).

Coupling weather and fatigue into P_{comms} (Eq. 5) therefore both raises preparedness and, under shore-comms stress, converts that lever into fewer collisions and fewer unrecovered MOB fatalities. Storm Corridor remains a hard environment for fatal method separation because environmental radio blackout is shared; Deep Water survival complements move directionally but are not primary claims. Future work should stress multi-ship COLREGs arbitration and heterogeneous manning.

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A. Default Constants

Table 8 lists principal defaults (overridable in configuration). Scenario overrides that differ from these defaults appear in Table 2.

Weather-field process (Mixed preset): regime flips $p_{cs} = 0.01$, $p_{sc} = 0.027$ per 5 min tick; Calm/Storm reversion targets $(0.30, 0.80, 0.25)$ / $(0.72, 0.30, 0.75)$; up to 8 storm cells; AR(1) coefficients $(\rho_s, \rho_v, \rho_u) \approx (0.983, 0.965, 0.976)$, $(\sigma_s, \sigma_v, \sigma_u) = (0.07, 0.06, 0.07)$; gradient cap 0.12. Calm and Stormy presets scale the flip rates.

B. Playback Workbench

The optional React 19 / Vite workbench is a batch-first client for the Axum API: it does not run the ABM in the browser. Figure 11 shows the shell—a permanent left panel beside a full-bleed deck.gl map over a MapLibre basemap. Under RUNS, the operator selects scenarios and methods, seed/tick counts, and an optional JSON config override, then posts a factorial batch; progress arrives over SSE while finished seeds appear in a completed-run list. Selecting a seed loads its JSONL tick log and enables the pinned playback dock (play/pause, speed, scrubber). Map layers render vessels and ports, AIS route context, weather/storm overlays, collisions and wrecks, and SAR/MOB entities, with camera and layer toggles in the map chrome.

Figure 12 shows the COMMS tab: bridge messages up to the scrubber (collision warnings with CPA/TTA, give-way, resume-route, keep-distance). STATISTICS and TELEMETRY (not shown) pull pairwise Mann–Whitney / Holm summaries and KPI time series for the active batch. The GUI is for inspection and debugging—all hypothesis numbers in the main text come from exported batch JSON / SQLite, not from the browser.

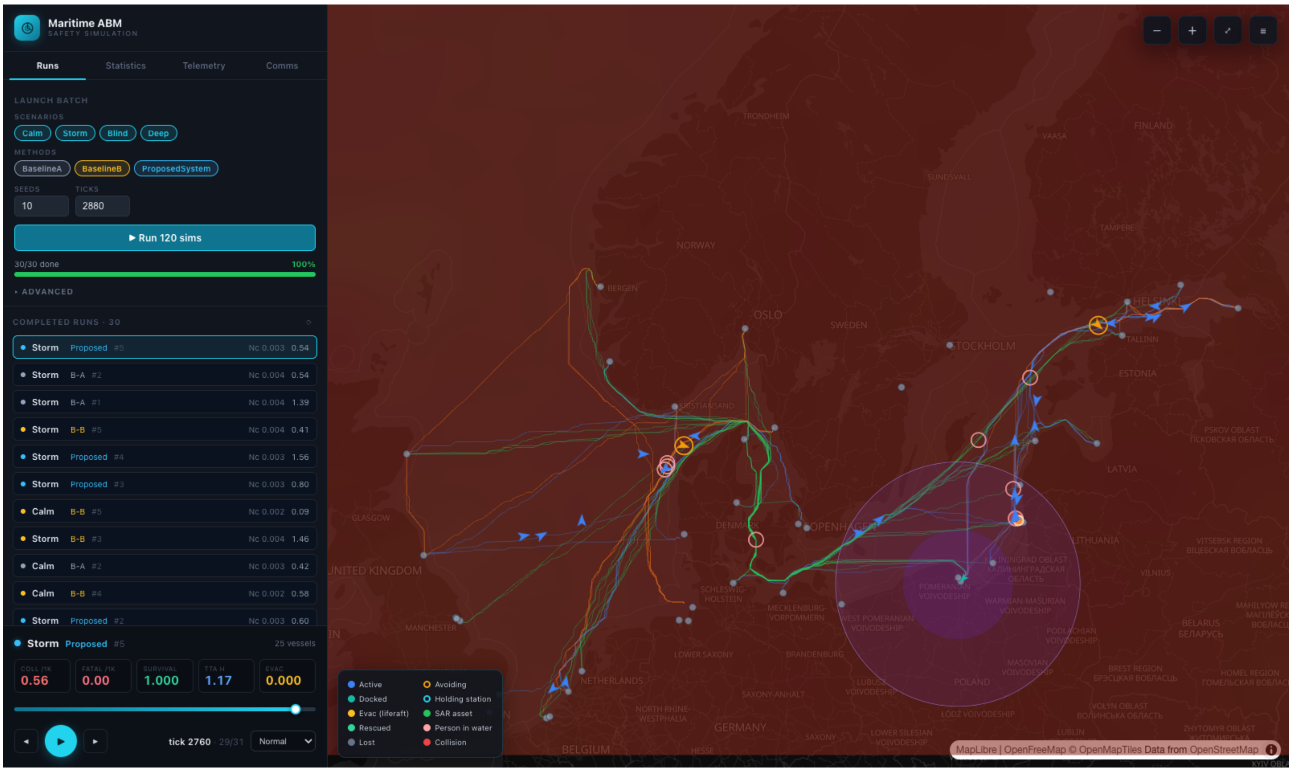

Figure 11: *Workbench shell scrubbed mid-run (Storm Corridor, Proposed, seed 5, tick 2760): live KPI tiles, vessel tracks / avoidance rings / collisions on the map, and the storm disc over the southern Baltic.*

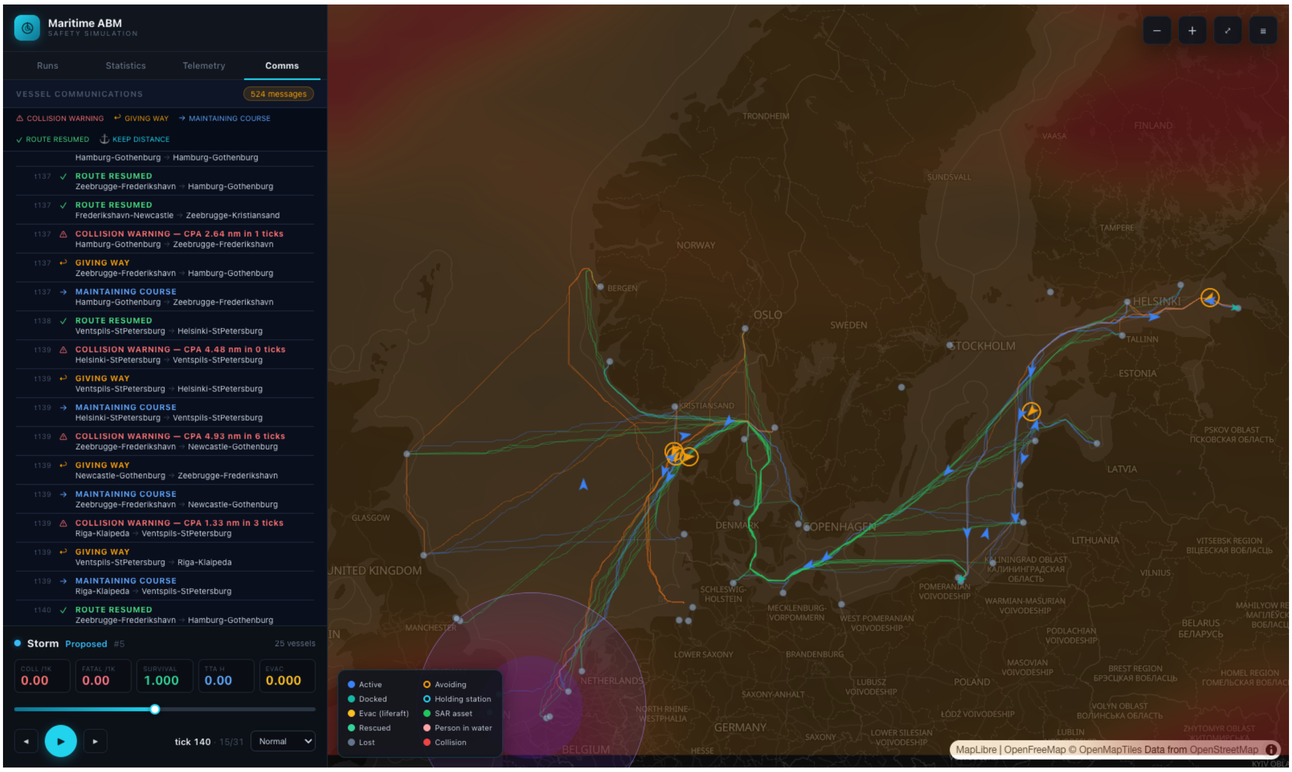

Figure 12: *Comms log for the same Storm Corridor run (hundreds of VHF-style messages with CPA warnings and give-way acknowledgements), with the playback map still showing routes and the moving storm.*

Table 8: *Selected engine default constants (role and value rationale). Scenario overrides appear in Table 2; longer discussion in Section 8.2.*

Constant	Default	Role / rationale
$\Delta t, T$	5 min; 14/4/7 d	Tick; Calm / Storm / Blind·Deep horizons (exposure vs event)
N vessels	750	Concurrent *underway* agents: Baltic AIS mean ≈ 774 moving ships [28] (HELCOM $\approx$2000 total at sea [27]); Storm/Blind 900, Deep Water 500. Stationary/moored omitted.
n crew	20	Fleet-mean overboard exposure; IMO A.1047 leaves manning to Administrations [29]; cargo/tanker practice $\approx$15–25 [27].
weather grid; (w_s, w_v, w_u)	50×50; $(1.0, 0.5, 0.8)$	Spatially resolved hazard for W; sea-state weighted highest as the dominant radio/perception stressor.
q (nominal link)	1.0	Perfect fair-weather VHF/AIS baseline so Blind Shore (0.55) and storm-comms isolate degradation.
warn / CPA / TTA gate	25 / 8 nm / 48 tk	Early COLREGs-style detection ($\approx$4 h); wider CPA berth at concurrent density [10, 11].
L_f, L_o; cooldown	5 / 9 nm; 2 tk	Larger starboard give-way offset (Eq. 6); $\approx$10 min lockout at $\Delta t = 5$ min.
waypoint reach	0.5 nm	Capture radius $\approx$ ship-length scale so pursuit does not chatter at each vertex.
r_{coll}; port exclusion	0.012 nm; 2.5 nm	Hard hit at $2r$=0.024 nm ($\approx$45 m, beam-scale) so Calm A stays in 0.5–2.0/ 1k ship-hrs at $\Delta t = 5$ min; harbour water excluded.
follow vicinity / gap; same-dest.	3 / 0.5 nm; 5 nm	Queueing on shared lanes: speed-match and hold-station without side-by-side overlap the give-way rule misses.
dwell	8–48 tk	Uniform 2–12 h port stay spreads departures and avoids convoy synchrony.
m(A/B/Proposed)	1.00/0.82/0.65	Method lever on Eq. 3: unmanaged / shore-rest / smart watch (H1–H3 ordering).
$(\beta_{\text{base}}, \beta_W, \beta_c)$	0.0015, 0.0040, 0.0008	Tuned so F rises over a watch with weather and circadian extras [14], not saturating in one tick.
recovery; F^*, κ	0.025/tk; 0.40, 0.65	Dock rest clears fatigue in $\approx$1 d; knee/penalty shape $g(F)$ so only elevated F cuts P_{comms}.
α_M; fatigue weight	2.0/1.3/1.0; 0.70	Awareness weights for P_{prep} (A most weather-sensitive); 0.70 keeps fatigue material beside W.
f_W (B, Proposed)	$\max(0.30,\ 1 - 0.45\,W)$	Weather-aware slowdown floor 0.30; Baseline A ignores (no mitigation).
storm radius / drift	120 nm / $\approx$1.0 nm/tk	Corridor: 160 nm cell at $\approx$12 kn ($\approx$3 d Channel$\to$Gdańsk).
storm-comms	0.30 (0.08 Corridor)	Environmental blackout shared by all methods; Corridor near-blackout (exploratory vs Blind H1/H3).
storm speed factor	0.70	Weather-aware crawl; A stays 1.0; Corridor non-A 0.45 for stronger Storm/Calm contrast.
force-field track	off	Optional Xiao-style repulsion [21]; off keeps discrete give-way as the primary COLREGs path.
$(m_{\text{sway}}, m_{\text{surge}})$	0.85, 0.05	DTU SIMCOL added-mass defaults [2].
Minorsky a, b	47.2, 32.7	Classical energy–volume coefficients (MJ, m^3) [23].
$\ell/L_{pp}, \tau$	0.12, 0.08 m	Damage-length fraction and side-shell thickness in the SIMCOL penetration surrogate.
$r_0, r_1, S^\dagger$	0.10, 0.30, 0.35	SOLAS-shaped S_i knees [24]; $S^\dagger$=0.35 keeps moderate damage afloat $\to$ MOB path (KPI 1).
helo / patrol; cutoff	80 / 25 kn; 40 nm	Typical SAR asset speeds; 40 nm selection range (Deep Water slows assets; Section 8.2).
Δt_{mob}; arrival r	6 tk; 1 nm	30 min readiness ($\approx$1 h Deep); datum capture radius before search starts.
w, r_0^{SAR}, u_d	5, 2 nm, $\approx$0.067 nm/tk	MASSIM/Koopman sweep width, initial search radius, and $\approx$0.8 kn drift [5, 6].
survival / τ_{mob}	288 / 18 tk	Liferaft 24 h; MOB $\approx$1.5 h in colder IAMSAR immersion bands [4, 31].
MOB scatter; searchers	0.5 nm; 1	Compact person-in-water cloud; one search team keeps rescue latency on the critical path.
sim. month	July (Ashrafi A); Deep Water Jan (D)	Mild default vs winter Group D degradation (exploratory Deep arm) [7].
P_c (IWRAP)	0.5–1.3$\times 10^{-4}$	IALA IWRAP Mk II causation defaults (head-on / overtaking / crossing) [3].
route weights (w_h, w_d, w_ℓ)	$(3, 4, 2.5)$	Offline A* prefers depth and lane density over raw length (Section 3.3).
UKC buffer m_s	2 m	Extra navigable margin beyond charted UKC for offline bathymetry routing.
draft+UKC (pax/cargo/tanker)	6.5+1.5 / 11+2 / 14+3 m	Baltic–North Sea class drafts: Ro-Pax $\approx$6.4 m [30]; cargo/tanker under PIANC [19].